

Data Science – Project Guidelines

A project will start with data collection followed by data preparation. Next, the project has two significant aspects as follows and a group must perform both.

1. Get inferences from the data using statistical analysis such as hypothesis testing. Know your tests and the reason for using them for your problem. Conduct a comprehensive comparison with other tests.
2. Train and validate some ML models (regression, classification, clustering, decision tree etc.) on the data set. Know your ML model and the reason for using it for your problem. If it is possible to apply other models, then compare your model comprehensively with others.

The guidelines for your data science projects are as follows:

- 1) If you have your data, then go to step 2; else, collect your data from the [repository](#)—preferably mixed data type of size at least 3M (instance x attribute) on which you will train a model.
- 2) Understand your data. For example, you should understand various features of the dataset, responses (if available). Formulate a problem statement that is meaningful, e.g., something that relates to your feature set, response etc.
- 3) Prepare your data (encoding, handling outliers, missing values etc).
- 4) Do preliminary EDA on the dataset as much as needed to understand more about the data.
- 5) Write a report in Google Docs detailing where you have collected the dataset,
 - a) how many rows and columns there are, and name some features with their interpretation to the real world.
 - b) How did you prepare your data, and why do you follow such steps?
 - c) What are the changes you got after data preparation?
 - d) What are your basic questions about your data, and how your problem statement is novel from that?
 - e) Based on your problem statement, what are your statistical inferences?
 - f) The report should contain all the codes written by you, a readme file.
- 6) You have to submit this report by mid October. A specific deadline will be announced later in the classroom.
- 7) This will be followed by a presentation, whose details will be announced during report submission deadline.

In general for this task, you will be graded based on how much you could dig into and infer from the data.

All the best!